

Too Far to Matter? The Urban Reach of Rural Territorial Conflict

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Can the economic reach of persistent territorial conflict extend to nearby urban markets even when conflict events occur outside the city and formal institutions remain operative? I study this question in southern Chile, where recurrent conflict over Mapuche territorial claims is predominantly rural. I link geocoded conflict events to a panel of 104 fixed urban markets and measure annual external exposure from attacks, coercive actions, and land invasions occurring within 30 kilometers but outside the receiving market. Greater external conflict exposure is associated with fewer registered enterprises and lower tax-reported sales. Moving from the 25th to the 75th percentile of positive exposure is associated with 7.05 percent lower sales. The decline in registered enterprises is accompanied by fewer first registrations, while recorded terminations show little comparable increase. The directly spatial urban outcomes do not display a common contemporaneous decline. The formal-activity association is also not systematically stronger in markets with greater historical *Títulos de Merced* coverage or larger baseline Mapuche population shares. The results therefore point to an urban economic reach of recurrent territorial conflict that is more visible in formal business activity than in a common contemporaneous physical pattern, and that should not be interpreted as an effect of Indigenous presence itself. The findings indicate that the economic implications of unresolved territorial conflict may extend beyond contested land even where formal institutions remain operative.

Keywords: Indigenous conflict; political violence; local economic activity; spatial spillovers; Chile.

JEL codes: D74; O18; R11.

1 Introduction

Conflict is usually measured where violence occurs, but its economic costs may extend to the towns and cities that organize trade, services, and formal business activity for the surrounding territory. When recurrent conflict in a rural hinterland changes expectations about security, market access, or the future use of surrounding land, firms may adjust even if the urban market itself is not attacked. This paper asks whether persistent territorial conflict reaches nearby urban markets and is associated with weaker formal economic activity.

Existing work shows that conflict affects firms, investment, and asset values, but it largely studies direct exposure to violence. Armed conflict increases plant exit and reduces sales, employment, and input use, while insecurity is capitalized into property and financial assets (Camacho and Rodriguez, 2013; Del Prete et al., 2023; Besley and Mueller, 2012; Abadie and Dermisi, 2008; Zussman et al., 2008). A smaller literature shows that conflict can reach other places through transport disruption or changes in perceived risk (Alfano and Cornelissen, 2026; Guidolin and La Ferrara, 2010; Brancati et al., 2026; Galdo et al., 2020). What remains less understood is whether persistent, low-intensity territorial conflict extends beyond event locations into nearby urban markets when formal state and market institutions continue to operate.

The closest evidence from southern Chile documents regional economic losses associated with violence (Larroulet Vignau and Sepúlveda, 2025); this paper goes further by looking inside the regions. Southern Chile provides a distinctive setting for this question. The conflict between the Chilean state and Mapuche organizations concerns Indigenous self-determination, territorial claims, restitution, and control over land rather than conventional civil war (Cayul et al., 2022). Its current geography reflects the late-nineteenth-century incorporation of Mapuche territory, the fragmentation of recognized land claims, and later cycles of land transfer and restitution (Jaimovich and Jordán, 2025b,a; Jaimovich and Toledo-Concha, 2026). The conflict studied here is predominantly rural. Of the 2,775 attacks, coercive actions, and land invasions used in the main event family, 82.5 percent lie outside the fixed urban systems used to construct the empirical exposure measure. Yet property registration, taxation, Indigenous land rules, and restitution remain governed by functioning formal institutions. This combination allows the paper to study the urban reach of persistent rural territorial conflict without the generalized institutional breakdown typical of conventional armed conflict.

The analysis follows 104 urban markets in Biobío, Araucanía, Los Ríos, and Los Lagos. The corridor is selected for substantive rather than nationally representative reasons: it covers the part of south-central Chile where historical Mapuche and Huilliche territorial occupation, historically recognized claims, contemporary restitution, and the recent geography of conflict most clearly overlap (Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas, 2008; Jaimovich and Jordán, 2025b; Jaimovich et al., 2025; Cayul et al., 2022).

I link annual geocoded events from the Mapuche Conflict Event and Data Archive (MACEDA) to each market. The main measure of external conflict exposure counts attacks, coercive actions, and land invasions occurring within 30 kilometers but outside the receiving urban market, while own-market conflict is controlled for separately. The main outcomes are registered enterprises and tax-reported sales from Chile’s national tax authority (SII). I complement these measures of formal activity with urban nighttime luminosity, fiscal appraisal, and newly built area to examine whether changes in formal activity are accompanied by broader urban physical adjustment.

Greater external conflict exposure is associated with fewer registered enterprises and lower tax-reported sales. Moving from the 25th to the 75th percentile of positive exposure is associated with 7.05 percent lower sales. The decline in registered enterprises is accompanied by fewer first

registrations, while recorded terminations show little comparable increase. The directly spatial outcomes do not display a common contemporaneous decline. Fiscal appraisal is close to zero contemporaneously but shows suggestive declines at medium lags, consistent with its interpretation as a slowly updated assessed stock rather than a high-frequency market price. The results therefore point more clearly to adjustment in formal economic activity than to a common contemporaneous physical pattern.

The historical geography provides an additional distinction. The negative formal-activity association does not become systematically stronger either where historical *Títulos de Merced* cover a larger share of the surrounding territory or where the baseline Mapuche population share is higher. This does not imply that historical claims or Indigenous presence are economically irrelevant, but it provides no evidence that either mechanically generates the negative association with contemporary conflict exposure.

The paper makes three contributions. First, it documents an overlooked spatial margin in the economics of conflict by tracing predominantly rural territorial conflict into nearby urban markets. Second, it distinguishes adjustment in formal business activity from broader contemporaneous changes in the urban physical environment. Finally, it shows that this spatial reach can emerge in an Indigenous self-determination conflict that is persistent and predominantly rural while property, tax, land-registration, and restitution institutions remain operative.

2 Institutional Setting and Conflict Geography

2.1 Territorial self-determination and historical claims

The Mapuche are an Indigenous people of south-central Chile whose historical territory encompasses much of the study corridor. Mapuche communities retained substantial autonomy through most of the colonial period and were incorporated into the Chilean state only after the military occupation of Araucanía in the late nineteenth century. Between 1884 and 1929, the state confined Mapuche communities to nearly 3,000 reservations covering roughly 550,000 hectares, a small share of their ancestral territory (Jaimovich and Jordán, 2025b; Jaimovich and Toledo-Concha, 2026). The resulting *Títulos de Merced* (TDM) created a state-recognized geography of claims that later land protection, restitution, and conflict continued to reference.

TDM boundaries do not describe current ownership. Allotment, land sales, weak enforcement, and later deregulated transfers reduced Mapuche ownership within many reservations (Jaimovich and Jordán, 2025b,a). Their empirical role is therefore to locate historically recognized claims, not to classify the current owner of a parcel or the ethnic identity of a nearby firm.

Agrarian reform extended the contested geography beyond the reservations. The reform began in 1962 and accelerated between 1970 and 1973, when Mapuche communities recovered or gained access to land through the expropriation of large estates. After the 1973 military coup, the dictatorship returned or reassigned much of that land, promoted forestry plantations, and divided remaining collective reservations into individual parcels. Contemporary invasions and attacks are

more common on plots touched by that reform and its reversal, linking current conflict to grievances both within and beyond TDM boundaries (Jaimovich and Jordán, 2025a; Jaimovich and Toledo-Concha, 2026).

2.2 Formal land institutions and restitution

The return to democracy placed these claims within a new formal legal system. The 1993 Indigenous Law recognizes Indigenous land through legal history, including TDM, long-standing occupation, judicial recognition, and land later transferred by the state. It restricts alienation and encumbrance, but the rules differ between community and individually held land and permit specified transactions with CONADI authorization (Gobierno de Chile, 1993). The law also created the Public Registry of Indigenous Territories, which makes registration central to the enforcement of Indigenous-land status. Evidence from the Temuco property records shows that registered parcels experienced fewer post-inheritance sales and smaller losses of Mapuche ownership, while registry coverage declined over time (Jordán et al., 2025).

CONADI administers the Fund for Indigenous Land and Water, through which the state has returned more than 200,000 hectares to Mapuche communities since the mid-1990s (Jaimovich et al., 2025). Restitution is administered through a formal process through which CONADI acquires and transfers land associated with recognized Indigenous claims. In the paper’s 104 markets, greater external conflict exposure is followed by more recorded CONADI land-acquisition events and acquired area over the subsequent two years. This illustrates the institutional setting of an active state.

Restrictions on transfer, collateralization, and use can affect investment and land markets, while later changes in enforcement can alter the economic salience of older claims (Field, 2007; Dippel et al., 2020; Feir et al., 2023).

2.3 Conflict geography and recurrence

The Mapuche Conflict Event and Data Archive (MACEDA) records geocoded events from local-media reports in the Mapuche self-determination conflict (Cayul et al., 2022). The main event family used in this paper contains attacks, coercive actions, and land invasions. Protests are excluded from the main exposure and retained as a separate event family. The events recur over time rather than defining a single absorbing transition into conflict.

The distinction between the setting’s rural geography and the empirical exposure matters. Among the 2,775 main events in the four-region corridor, 2,288 (82.5%) lie outside all fixed urban systems used in the exposure construction. The conflict studied here is therefore predominantly rural. The main exposure nevertheless remains broad external conflict rather than a measure restricted to rural-coded events. It counts eligible events outside the receiving fixed urban system, including events inside another urban market; Section 3.2 provides the exact spatial rule.

Figure 1 locates the four-region corridor and the distribution of main conflict events. The sample contains 104 markets in the area where historically recognized Mapuche and Huilliche

claims, contemporary restitution, and the geography of conflict most clearly overlap. La Araucanía contains 66.6 percent of the main events in the corridor despite accounting for 25.7 percent of its 2017 population, but exposure also extends into Biobío, Los Ríos, and Los Lagos.

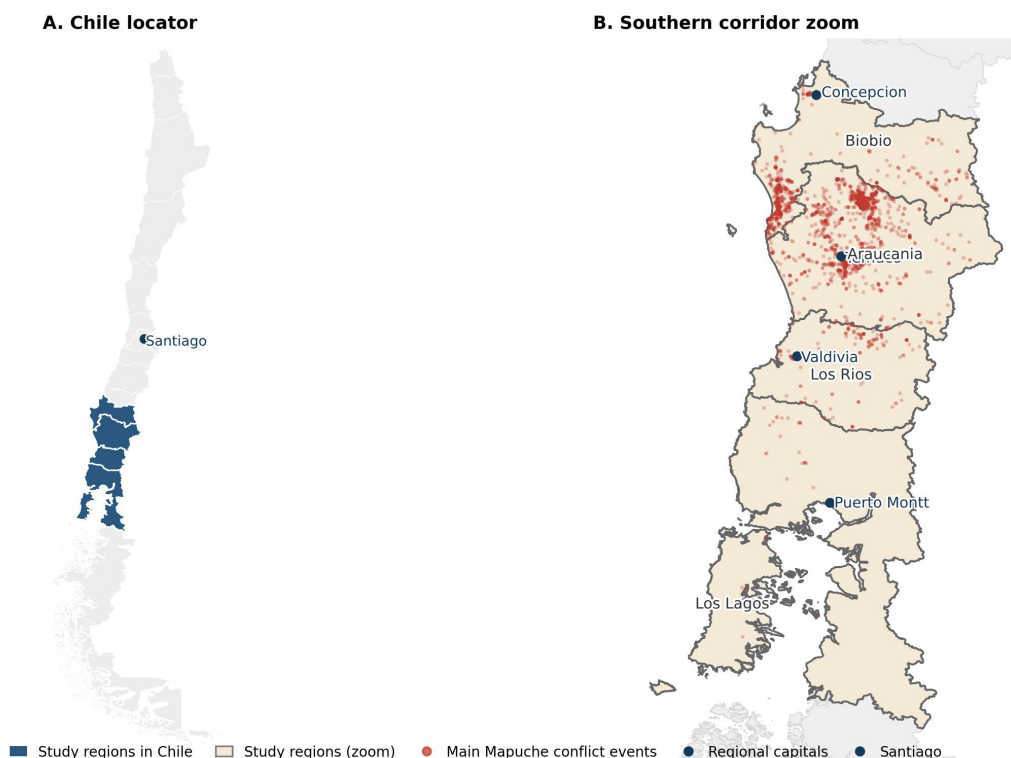


Figure 1: Study corridor in southern Chile. Panel A locates Biobío, Araucanía, Los Ríos, and Los Lagos within Chile and marks Santiago for orientation. Panel B displays the geography of attacks, coercive actions, and land invasions in the corridor.

Figure 2 shows the recurrent annual path and regional concentration of these events. The series rises and falls rather than moving once from peace to conflict. Nearby urban markets may therefore face repeated changes in external exposure even when events occur beyond their own urban footprint.

3 Data & Empirical Strategy

3.1 Units, outcomes, and measurement

The empirical analysis follows 104 fixed urban markets in Biobío, Araucanía, Los Ríos, and Los Lagos. Each market is defined from the urban *manzanas* in the 2017 INE geography and remains fixed over time. The urban market is the economic object of the paper, but the geographic support of the outcomes differs across data sources.

The outcomes capture different dimensions of formal economic activity and urban physical conditions. I begin with formal business activity, using annual administrative records from Chile's

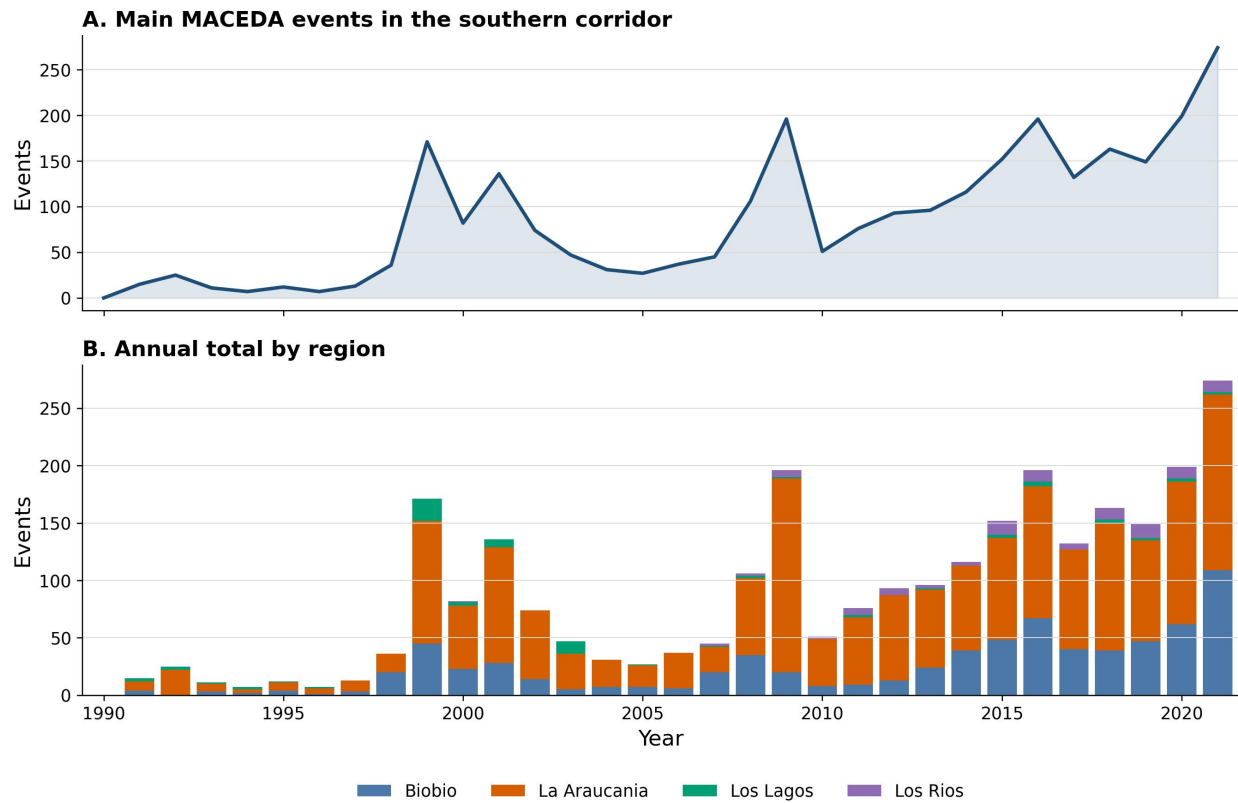


Figure 2: Main Mapuche conflict events in the four-region corridor. Panel A reports the annual count of attacks, coercive actions, and land invasions. Panel B reports annual counts separately for Biobío, Araucanía, Los Ríos, and Los Lagos.

national tax authority (Servicio de Impuestos Internos, SII). Registered enterprises measure the stock of firms reporting activity in each municipality, while tax-reported sales capture the scale of formal market activity. Both series are available annually from 2005 to 2021. Because SII assigns firms to the municipality of their tax or headquarters address, these outcomes measure formal activity associated with each urban market, but they do not by themselves establish that the underlying activity occurs within the fixed urban footprint.¹

Nighttime luminosity provides a directly spatial measure of activity within the fixed urban footprints. Harmonized satellite imagery allows me to follow the same urban polygons over a longer period than the SII business records. Lights do not measure formal firms or sales directly, but they provide a complementary indicator of whether greater conflict exposure is accompanied by a broader change in urban economic activity.

The property cadastre captures a slower-moving margin. SII records allow me to measure both the total fiscal appraisal of properties within the fixed urban geography and mean appraisal per property. Fiscal appraisal is the administrative value used for the property-tax base rather than a transaction price. It changes through mass revaluations, indexation, and updates to the registered property stock (Servicio de Impuestos Internos, 2006, 2026a,b). I therefore interpret it as a slowly updated measure of the assessed urban property stock. This distinction is important when interpreting movements that appear only at longer lags.

Finally, built-up-area data provide a more direct measure of physical urban development. WSF Evolution records changes in built surface rather than business activity or property values. I use it as a complementary measure of whether conflict exposure is associated with slower physical expansion of the urban footprint; because its temporal support and interpretation differ from the headline outcomes, the detailed results are reported in the appendix.

Together, these outcomes move from relatively responsive measures of formal economic activity to progressively slower and more physical margins of urban adjustment. Appendix A reports the exact construction, transformations, temporal coverage, and spatial support of each series.

3.2 External conflict exposure

Conflict exposure comes from annual geocoded events in the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022). The event family used in the main analysis includes attacks, coercive actions, and land invasions, while protests are excluded. Let A_{mt} denote the number of eligible events occurring within 30 kilometers of urban market m in year t but outside the receiving market. External conflict exposure is defined as

$$D_{mt} = \log(1 + A_{mt}). \quad (1)$$

¹Appendix E examines this spatial interpretation in two ways. Sectoral SII results and input-output measures of rural linkages show no disproportionate concentration of the aggregate association in sectors more structurally linked to rural activity. Separately, historical headquarters of a selected subset of enterprises can be linked with high confidence to the fixed urban footprints; the association remains negative within this urban-headquarters subset.

The count includes events occurring outside all fixed urban systems and events occurring inside a different fixed urban market, provided that they lie within 30 kilometers of the receiving market. The measure is therefore not restricted to rural-coded events. Although the conflict studied here is predominantly rural, what defines D_{mt} empirically is that the event occurs outside the receiving urban market.

The external count can be written as $A_{mt} = R_{mt} + X_{mt}$, where R_{mt} counts events outside every fixed urban system (purely rural) and X_{mt} counts events inside a different fixed urban market. Events in R_{mt} account for 79.6 percent of receiver–event pairs. When the two components enter jointly, the outside-system component is negatively associated with both registered enterprises and tax-reported sales. Appendix A reports the component estimates; the preferred exposure remains their combined count.

Main MACEDA events occurring inside the receiving market enter separately as $I_{mt} = \log(1 + B_{mt})$, where B_{mt} is the annual own-market count. This allows the external-exposure coefficient to be estimated conditional on contemporaneous conflict within the market itself. Appendix A reports the event-composition, support, and spatial-assignment rules, while Appendix B examines alternative distance definitions.

3.3 Empirical specification and identification

For outcome Y_{mt} , the preferred market-level specification is

$$Y_{mt} = \beta D_{mt} + \rho I_{mt} + \mu_m + \lambda_t + \tau_t(X_m^0)' \theta + \kappa \tau_t C_m + \varepsilon_{mt}, \quad (2)$$

where μ_m and λ_t are market and year fixed effects, respectively, and τ_t is a linear calendar-time index. The vector X_m^0 contains baseline characteristics, while C_m indicates capital status. Standard errors are clustered by urban market.

Market fixed effects absorb persistent differences across markets, while year fixed effects capture shocks common to the study corridor. Interactions between baseline characteristics and linear time allow markets with different initial demographic, economic, geographic, and conflict conditions to follow different trends. All variables are reported in Appendix B. The specification does not include contemporaneous economic or land-use controls that may themselves respond to conflict.

The coefficient β is estimated from within-market changes in external conflict exposure, relative to common annual shocks and the differential trends allowed by the baseline characteristics. I interpret β as a conditional association rather than as an effect identified from exogenous changes in conflict exposure.

A complementary exercise examines whether recent economic changes predict subsequent conflict exposure. One-year changes in enterprises and sales have little predictive power for exposure one or two years ahead, whereas persistently lower levels of formal activity remain associated with later exposure even after recent conflict history is taken into account. This offers little support for a simple short-run feedback channel from recent economic deterioration to subsequent conflict,

but it leaves open slower-moving local conditions, persistent territorial differences, longer conflict histories, and longer-run feedback. Appendix C reports the full estimates.

Appendix D complements the market-level analysis with a within-market spatial design using *manzana* fixed effects and market-by-year fixed effects. Because aggregate market-year changes are absorbed, this exercise captures relative spatial incidence within urban markets rather than aggregate changes in market activity.

3.4 Timing patterns

The timing analysis extends the preferred specification by including future, contemporaneous, and lagged values of the same annual exposure measure:

$$Y_{mt} = \sum_{h=1}^3 \pi_h D_{m,t+h} + \sum_{\ell=0}^{L_Y} \beta_\ell D_{m,t-\ell} + \rho I_{mt} + \mu_m + \lambda_t + \tau_t (X_m^0)' \theta + \kappa \tau_t C_m + \varepsilon_{mt}. \quad (3)$$

The preferred projection includes three future-exposure terms and lags through eight years for enterprises, sales, and fiscal appraisal. The nighttime-lights profile extends through ten years because of its longer temporal support.

For presentation, coefficients on future exposure are plotted at negative horizons. These terms show whether outcomes tend to move before the corresponding exposure is realized, while the contemporaneous and lagged coefficients trace how the association evolves during and after exposure. Because conflict exposure is continuous and recurrent, the exercise is best read as a description of its temporal pattern rather than as a conventional event study.

Appendix C reports additional timing evidence.

4 Main Results

4.1 Main results

Greater external conflict exposure is associated with weaker formal business activity in nearby urban markets. Registered enterprise counts are lower in more exposed markets, and tax-reported sales move in the same direction (Table 1). These associations are already visible with market and year fixed effects and remain after allowing trends to vary with predetermined baseline conditions, adding the capital-status trend, and conditioning on contemporaneous conflict inside the receiving market. In the final specification, the enterprise coefficient is -0.0112 and the sales coefficient is -0.0475 . Moving from the 25th to the 75th percentile of positive external exposure is associated with 7.05 percent lower tax-reported sales.

Most of the change across specifications occurs when the predetermined baseline-condition trends are introduced. Adding the capital-status trend and own-market conflict subsequently

Table 1: External conflict exposure and urban-market outcomes

	Formal economic activity		Urban physical outcomes		
	(1) Registered enterprises	(2) Tax-reported sales	(3) Nighttime lights	(4) Fiscal appraisal, total	(5) Fiscal appraisal, mean
Market FE + year FE	-0.024*** (0.009)	-0.063*** (0.021)	-0.008 (0.011)	-0.012 (0.023)	-0.005 (0.022)
+ predetermined baseline trends	-0.011** (0.005)	-0.046*** (0.015)	0.008 (0.009)	-0.000 (0.016)	-0.002 (0.015)
+ capital-status trend	-0.012** (0.005)	-0.047*** (0.015)	0.009 (0.009)	0.002 (0.016)	-0.001 (0.015)
+ contemporaneous own-market conflict	-0.011** (0.005)	-0.047*** (0.015)	0.007 (0.009)	0.002 (0.016)	-0.000 (0.015)
Observations	1,768	1,768	3,060	1,248	1,248
Markets	104	104	104	104	104
Sample period	2005–2021	2005–2021	1992–2021	2010–2021	2010–2021

Notes: Each cell reports the coefficient on log broad external conflict exposure within 30 kilometers; market-clustered standard errors appear below. The formal outcomes and directly spatial stock measures enter in logs. Every specification includes market and year fixed effects. Rows sequentially add predetermined baseline-condition trends, a capital-status trend, and contemporaneous log conflict inside the receiving market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

changes the enterprise and sales estimates little. The final addition of own-market conflict changes both coefficients only slightly.

The decline in formal business activity does not extend to every urban outcome. In Table 1, nighttime lights do not fall with exposure, and both total and mean fiscal appraisal remain close to zero. The evidence therefore points to weaker formal business activity, not to an immediate decline across every measure of urban development. Fiscal appraisal is an administratively updated stock rather than a transaction price, so it may respond only gradually to changes in local conditions.

Appendix B reports the corresponding estimates using cumulative exposure windows from 10 to 60 kilometers. The negative formal-activity association is not specific to the 30-kilometer window.

4.2 Timing patterns

Registered enterprises fall when nearby conflict exposure rises, and the decline becomes larger over the following years. Figure 3 does not show a comparable pattern before exposure. This timing is consistent with weaker formal business activity following nearby conflict, although it does not by itself establish that exposure is exogenous.

Tax-reported sales also fall when nearby conflict exposure rises. Yet sales are already lower before the measured rise in exposure, which makes their timing harder to interpret than the enterprise results.

Fiscal appraisal does not change at the time of exposure, but is lower at medium horizons. Because it also moves before current exposure, this pattern cannot be read as a delayed causal response. It is more consistent with an administratively updated property stock than with continuous

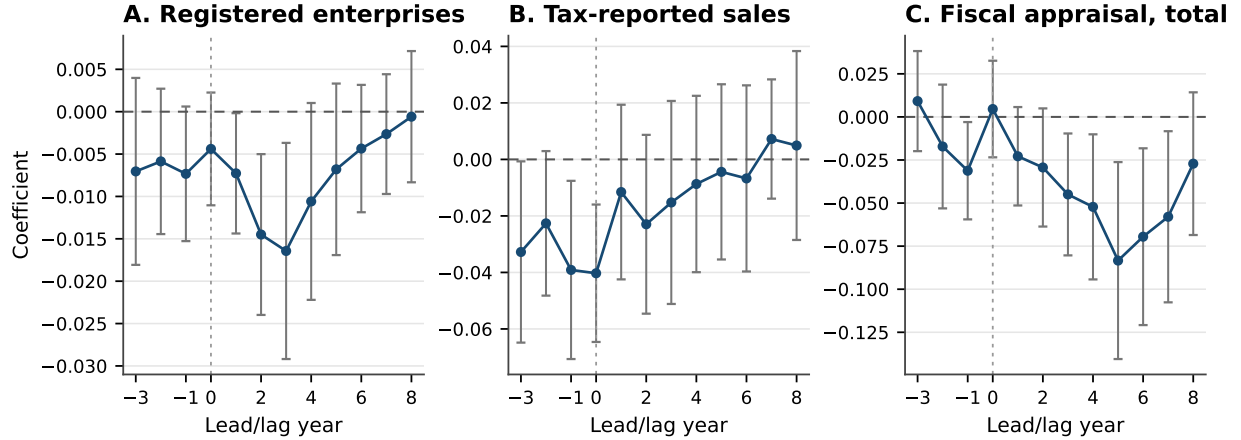


Figure 3: Timing patterns in formal activity and fiscal appraisal. The panels report coefficients from the continuous lead-lag projection defined in Section 3. Negative values on the horizontal axis are future exposure terms, zero is current exposure, and positive values are lagged exposure terms. Vertical bars are 95 percent confidence intervals based on standard errors clustered by urban market. All specifications use broad external exposure within 30 kilometers, condition on contemporaneous own-market conflict, and include market and year fixed effects, predetermined baseline-condition trends, and a capital-status trend.

market repricing.

A complementary non-absorbing difference-in-differences design centers the analysis on unusually large annual escalations in nearby external conflict. Registered enterprises are approximately flat during the three placebo years and become progressively lower after escalation under both the annual Barrett-style composite and the fully past-determined Past4SD rule. Tax-reported sales do not display the same stable dynamic profile. Appendix C.2 presents the treatment definitions, identifying comparison, and full estimates.

4.3 Spatial interpretation and supporting evidence

The SII outcomes are measured at the municipal tax address, so the aggregate estimates do not by themselves establish that the decline in formal activity occurs inside the fixed urban footprint. Two additional exercises help assess this interpretation. For a selected set of registered legal entities (*personas jurídicas*), historical headquarters can be linked with high confidence to the property cadastre. Restricting this sample to headquarters located inside the fixed urban footprints produces a negative estimate of -0.0224 . The linked sample is selective, and headquarters need not coincide with production or sales locations, but the result shows that the negative association is also present among registered legal entities whose historical headquarters can be located inside the fixed urban footprints.

A complementary sectoral exercise asks whether the municipal result is driven mainly by activities that are structurally tied to the rural economy. The association is not systematically larger in sectors with stronger predetermined agricultural and forestry input or customer linkages. This

evidence does not locate SII activity within the urban footprint, but it makes an explanation based exclusively on rural-linked sectors less compelling. Appendix E reports both exercises.

The block-level analysis in Appendix D asks a different spatial question. It compares more and less exposed *manzanas* within the same urban market and year, using manzana fixed effects and market-by-year fixed effects. The estimates therefore capture relative differences across locations inside a market rather than aggregate changes in that market. Across nighttime lights, fiscal appraisal, new built area, and land-use composition, the existing estimates do not reveal a stable pattern of within-market spatial reallocation.

5 Additional Evidence

5.1 Robustness

The negative formal-activity coefficients remain when annual shocks are allowed to vary across regions or provinces. Replacing year fixed effects with region-by-year effects yields coefficients of -0.0170 for registered enterprises and -0.0526 for tax-reported sales. With province-by-year effects, the corresponding estimates are -0.0067 and -0.0398 . The enterprise estimate becomes less precise in the latter specification, but remains negative and similar in magnitude. The results are also not driven by any one province. Re-estimating the main specification after excluding each province in turn produces enterprise coefficients between -0.0135 and -0.0066 , and sales coefficients between -0.0530 and -0.0366 .

The choice of the 30-kilometer exposure window is not critical to this pattern. Using cumulative exposure within 10, 20, 30, 40, 50, or 60 kilometers produces negative coefficients for both enterprises and sales throughout. I interpret these estimates as sensitivity to the spatial window rather than as evidence for an optimal radius.

A separate concern is the importance of aquaculture in Los Lagos. Excluding SII subsector 032 (Acuicultura) leaves the enterprise coefficient almost unchanged, from -0.0112 to -0.0114 . Because suppression of subsector-level sales reduces the observable sample, the sales comparison is made on the same 1,066 market-years. On that sample, the coefficient changes from -0.0516 to -0.0550 after aquaculture is removed. The main formal-activity pattern therefore does not appear to be driven by the aquaculture sector. Inference is also similar under wild-cluster bootstrap and spatial-HAC standard errors that allow correlation across nearby markets and serial dependence within markets. Appendix B reports the full robustness results and the leave-one-province-out estimates.

Given the historical roots of the conflict, I also examine whether the formal-activity association varies with exposure to historical *Títulos de Merced* or with the Mapuche share of the population in 1992. Although the interactions are imprecisely estimated, the estimates provide no evidence that the association becomes systematically stronger with greater historical claim exposure or baseline Mapuche presence. Appendix F reports the joint estimates and multiplicity-adjusted inference.

5.2 Margins of adjustment

The decline in registered enterprises may reflect fewer new registrations, more terminations, or both. External conflict exposure is associated with fewer first registrations, with a non significant results for terminations. This suggests that fewer new registrations are one margin contributing to the decline in the enterprise stock. The evidence therefore suggests that fewer new registrations contribute to the enterprise decline. Appendix B reports the remaining firm-dynamics results.

6 Discussion and Conclusion

This paper asks whether the economic reach of persistent territorial conflict extends beyond the places where events occur to nearby urban markets, even when formal institutions remain operative. Using geocoded conflict events in southern Chile, I find that greater external conflict exposure is associated with fewer registered enterprises and lower tax-reported sales. Moving from the 25th to the 75th percentile of positive external exposure is associated with 7.05 percent lower sales. These associations remain after conditioning on contemporaneous conflict inside the receiving market and allowing trends to vary with predetermined local characteristics.

The results across outcomes suggest a particular pattern of adjustment. The decline in registered enterprises is accompanied by fewer first registrations, while recorded terminations show little comparable increase. On samples shared with each physical series, sales remains negatively associated with exposure, but the directly spatial measures do not show a common contemporaneous decline: nighttime lights are positive in their common SII window, whereas fiscal appraisal and newly built area are close to zero. Persistent uncertainty about security, market access, surrounding land use, or the future course of territorial disputes may make firms less willing to enter, register, or expand activity even when the urban market itself is not directly attacked. The data do not allow these channels to be separated, and changes in tax-reported activity need not map one-for-one into changes in production.

The spatial evidence also speaks against a purely rural-production explanation. The aggregate formal-activity association is not systematically stronger in sectors with greater predetermined agricultural and forestry linkages, and it remains negative among a selected group of registered legal entities whose historical headquarters can be located with high confidence inside the fixed urban footprints. The within-market evidence likewise does not reveal a broad negative spatial gradient in physical outcomes across more and less exposed blocks. Taken together, these results suggest that nearby territorial conflict can be associated with changes in the economic organization linked to an urban market without producing a corresponding contemporaneous decline across the urban physical landscape.

An important distinction concerns the historical and Indigenous geography of the conflict. Contemporary exposure is closely connected to a setting shaped by Mapuche territorial claims and historical dispossession, but the empirical association should not be interpreted as an effect of Indigenous presence itself. The negative formal-activity gradient does not become systematically

stronger either where historical *Títulos de Merced* cover a larger share of the surrounding territory or where the baseline Mapuche population share is higher. These results do not imply that Indigenous demographic presence or historical territorial claims are economically irrelevant. Rather, they provide no evidence that either mechanically generates the negative formal-activity association with contemporary conflict exposure.

These findings also have a spatial implication for policy. Assessing the local economic implications of territorial conflict only where events occur may overlook effects on nearby urban markets. The costs of recurrent territorial conflict may therefore extend to the nearby urban markets that organize formal activity for the surrounding territory.

The institutional evidence adds a further distinction. Conflict in southern Chile does not unfold in the absence of a functioning state. Property registration, taxation, Indigenous land protections, and restitution remain operative, and periods of greater external conflict exposure are followed by more recorded CONADI land-acquisition activity. Institutional presence, however, is not the same as institutional resolution. Recorded land-acquisition activity is higher following greater conflict exposure, but the analysis does not establish whether restitution subsequently mitigates or amplifies conflict or its economic consequences. The evidence therefore does not establish whether more or less restitution would reduce these costs. It does suggest that the economic stakes of how territorial claims are managed extend beyond the contested land itself: when disputes remain recurrent, their association with economic activity can reach the nearby urban markets that organize formal business activity for the surrounding territory.

The estimates are conditional associations rather than effects identified from an external source of variation in conflict. Persistent local conditions may shape both conflict and economic activity over longer horizons, and the headline SII outcomes measure formal activity associated with urban markets through municipal tax addresses. The evidence comes from a specific institutional and spatial setting: recurrent, relatively low-intensity territorial conflict connected to nearby service and trading centers, with functioning tax, property, land-registration, and restitution institutions. It therefore speaks most directly to settings with strong rural–urban economic linkages, rather than implying that rural conflict generally harms cities. Within those limits, greater nearby conflict exposure is associated with weaker formal activity in urban markets that are not themselves the main sites of confrontation.

The costs of unresolved territorial conflict may therefore extend beyond the actors and parcels directly involved to the urban markets that connect the rural hinterland to the wider economy.

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A Data, measurement, and sample construction

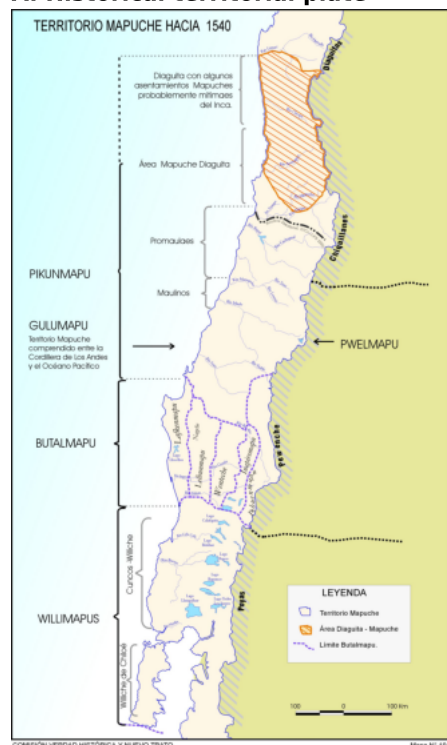
A.1 Urban-market geography and study sample

The analysis follows 104 urban markets in Biobío, Araucanía, Los Ríos, and Los Lagos. Each market is defined as the union of the urban *manzanas* assigned to its municipality in the 2017 INE geography and is held fixed over time. The four-region corridor is selected for substantive rather than nationally representative reasons: it covers the part of south-central Chile where historical Mapuche and Huilliche territorial occupation, recognized land claims, contemporary restitution, and the recent geography of conflict most clearly overlap.

These fixed urban markets define the spatial unit used to organize the analysis. SII enterprise and sales records are observed at municipal tax addresses and mapped to the corresponding urban market; they therefore measure formal activity associated with the market rather than activity mechanically located inside its urban footprint.

Figure A1 provides historical and demographic context for the study corridor.

A. Historical territorial plate



B. Contemporary corridor and 1992 Mapuche shares

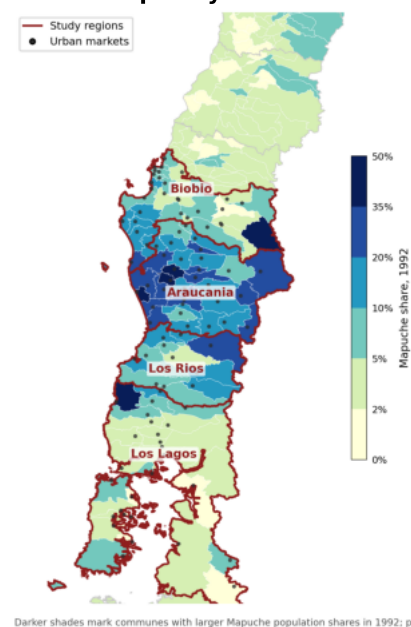


Figure A1: Historical Mapuche territory and the contemporary study corridor. Panel A reproduces a historical map of Mapuche territorial organization around 1540 from the Comisión Verdad Histórica y Nuevo Trato. Panel B shows the four-region study corridor and Mapuche population shares in 1992. Together, the panels illustrate the historical and demographic geography underlying the study area.

The business panel covers 2005–2021. Fiscal appraisal begins in 2010, while harmonized night-time lights and WSF built-area data have longer, but distinct, temporal support. Table A2 records the construction and transformation of each outcome used in the analysis.

A.2 Conflict data and event composition

Conflict events come from MACEDA. The main analysis counts attacks, state coercive actions, and land invasions; protests are excluded. This family contains 2,775 events, of which 82.1 percent have MACEDA's rural classification and 82.5 percent lie outside all fixed urban systems used to construct the exposure measure. The descriptive geography is thus predominantly rural, whereas exposure is defined by externality to the receiving market.

Table A1: MACEDA event composition and regional concentration

Event family	Events	Rural MACEDA	Share	
<i>Panel A. Event composition</i>				
All MACEDA events	3,730	2,587	69.4%	
Main event family: attacks, coercion, land invasions	2,775	2,278	82.1%	
Protests	955	309	32.4%	
<i>Panel B. Operational fixed-system geography of the main event family</i>				
Classification	Events	Share		
Outside all fixed urban systems	2,288	82.5%		
Inside a fixed urban system	487	17.5%		
<i>Panel C. Regional concentration of main events</i>				
Region	Main events	Event share	2017 population	Population share
Biobio	752	27.1%	1,556,805	41.8%
La Araucanía	1,847	66.6%	957,224	25.7%
Los Lagos	76	2.7%	828,708	22.2%
Los Ríos	100	3.6%	384,837	10.3%
<i>Panel D. La Araucanía share of cumulative main events</i>				
Window	Main events	La Araucanía events	Share	
1990–1992	40	30	75.0%	
1990–2002	589	401	68.1%	
1990–2017	1,990	1,371	68.9%	
1990–2021	2,775	1,847	66.6%	

Notes: The main event family comprises attacks, coercive actions, and land invasions. Rural is MACEDA’s event classification. The operational outside-system count uses the fixed urban systems. Protests are not used in the exposure measure.

A.3 External exposure construction

For market m and year t , external exposure is $D_{mt} = \log(1 + A_{mt})$, where A_{mt} counts eligible events within 30 kilometers of the receiving market and outside that market. It includes events outside all fixed urban systems and events inside another fixed urban market. Events inside the receiving market enter separately as annual own-market conflict, with no radius cutoff.

A.4 Outcome construction and SII measurement

Nighttime lights, fiscal appraisal, and WSF are extracted from fixed urban footprints. Fiscal appraisal is an assessed administrative stock, not a transaction price, and is updated more slowly than a high-frequency market price. WSF records newly built area and therefore measures a physical flow. For registered enterprises and tax-reported sales, the SII address is a tax or headquarters address; it does not identify where a firm produces or where its sales take place. Appendix E examines the implication of this distinction.

A.5 External exposure components

The external count combines events outside all fixed urban systems (R_{mt}) and events located inside a different fixed urban market (X_{mt}). Outside-system events account for 79.6 percent of receiver–event pairs. Table A3 enters the two components jointly while retaining the same fixed effects, trends, own-market co-exposure, and market clustering as the preferred formal-activity specification. The preferred measure remains $D_{mt} = \log(1 + R_{mt} + X_{mt})$. The imprecise estimates

External conflict exposure around selected urban markets

Events are counted within 30 km of the receiving market; protests are excluded

Temuco

Cañete

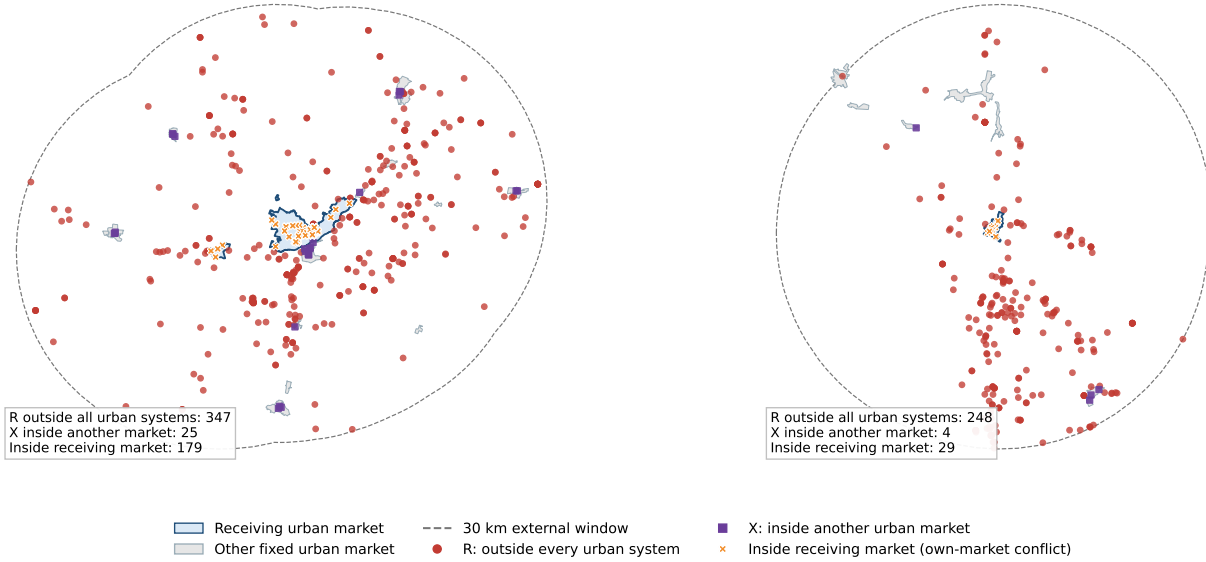


Figure A2: External conflict exposure around selected urban markets. The dashed boundary marks the 30-kilometer window around each receiving market. R events occur outside every fixed urban system; X events occur inside a different fixed urban market. Events inside the receiving market are shown separately as own-market conflict.

Table A2: Outcome construction and interpretation

Outcome	Source and spatial unit	Years	Transformation	Interpretation
Registered enterprises	SII municipality tax-address measure mapped to market	2005–2021	log(enterprises)	Formal enterprise stock associated with the market
Tax-reported sales	SII municipality tax-address measure mapped to market	2005–2021	log(sales)	Scale of formal activity associated with the market
Nighttime lights	Harmonized DMSP–VIIRS, fixed urban footprint	1992–2021	log(total DN)	Urban luminosity
Fiscal appraisal total	SII cadastre, fixed urban footprint	2010–2021	log(appraised value)	Assessed urban property stock
Fiscal appraisal mean	SII cadastre, fixed urban footprint	2010–2021	log(mean appraisal)	Mean assessed property value
WSF new built area	WSF Evolution, fixed urban footprint	1990–2015	asinh(new hectares)	New built-area flow

Notes: Registered enterprises and tax-reported sales are SII municipality tax-address measures mapped to urban markets. Nighttime lights, fiscal appraisal, and WSF are constructed within fixed urban footprints. Transformations follow each outcome's support and unit; they are not all logs.

for the X component do not establish the absence of urban-to-urban spillovers.

Table A3: Components of external conflict exposure

	Outside-system R		Other-market X		Total
<i>Panel A. Receiver-event pair support</i>					
Receiver-event pairs	9,114	2,331	11,445		
Share of pairs	79.6%	20.4%	100.0%		
Market-years with positive count	32.5%	20.3%	39.3%		
<i>Panel B. Joint-component specification</i>					
Outcome	β_R (SE)	p_R	β_X (SE)	p_X	$p(\beta_R = \beta_X)$
Registered enterprises	−0.0122** (0.0055)	0.029	0.0018 (0.0061)	0.764	0.088
Tax-reported sales	−0.0347** (0.0171)	0.045	−0.0270 (0.0169)	0.113	0.750
Corr[log(1 + R), log(1 + X)] = 0.493					

Notes: R denotes eligible events outside every fixed urban system within 30 kilometers of the receiving market; X denotes eligible events inside a different fixed urban market within the same window. Panel B enters log(1 + R_{mt}) and log(1 + X_{mt}) jointly in the preferred SII specification. Standard errors are clustered by market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.6 Exposure support

Table A4 describes the support for external and own-market exposure in the full 1990–2021 market panel. Own-market conflict captures a distinct contemporaneous geography from the external-exposure measure.

Table A4: Support for external and own-market conflict exposure

Exposure measure	Mean	SD	Positive market-years
External exposure, D_{mt}	0.690	1.050	39.3%
Own-market exposure, I_{mt}	0.063	0.283	5.9%
Corr(D_{mt}, I_{mt}) = 0.350			
Both positive: 5.0%; $D > 0, I = 0$: 34.3%; $I > 0, D = 0$: 0.9%			

Notes: $D_{mt} = \log(1 + A_{mt})$ is broad external exposure; I_{mt} is the log of one plus annual main conflict events inside the receiving market. The panel contains 104 markets from 1990 to 2021.

A.7 Matched formal and physical samples

The physical series differ in temporal support. Table A5 therefore re-estimates enterprises, sales, and each physical outcome on the same available market-years. Sales remains negative in every matched sample, while the enterprise estimates are negative but less precise in the shorter appraisal and WSF windows. Nighttime lights are positive in their common SII window, whereas fiscal appraisal and newly built area are close to zero.

Table A5: Formal activity and physical outcomes on matched samples

Physical series	Physical coefficient	Enterprises	Sales	Common sample
Urban nighttime lights	0.0269** (0.0103)	-0.0108** (0.0049)	-0.0474*** (0.0145)	1,761; 104; 17
Fiscal appraisal total	0.0057 (0.0156)	-0.0060 (0.0049)	-0.0374** (0.0166)	1,248; 104; 12
Fiscal appraisal mean	0.0010 (0.0142)	-0.0060 (0.0049)	-0.0374** (0.0166)	1,248; 104; 12
WSF new built area	-0.0107 (0.0320)	-0.0074 (0.0049)	-0.0403*** (0.0152)	1,144; 104; 11

Common sample reports market-years; markets; years.

Notes: Each physical outcome and both SII formal outcomes are estimated on the same rows for the indicated physical series. Coefficients are on external conflict exposure in the respective preferred specifications; standard errors are clustered by market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Robustness and margins of adjustment

B.1 Trend specification and sensitivity

The preferred formal-activity specification allows linear trends to differ with baseline market characteristics. Table A6 records the underlying variables and their timing. The capital-status trend is entered separately. Table A7 then reports one coherent sensitivity that removes the pre-period conflict, baseline Mapuche-share, and historical-title trends while retaining the other baseline-characteristic trends, capital-status trend, fixed effects, and own-market conflict. Sales is similar across the two trend bundles. The enterprise estimate remains negative but is less precise in the reduced specification.

Table A6: Baseline trends in the preferred formal-activity specification

Baseline characteristic	Reference period and source	Purpose of differential trend
Pre-period rural-only conflict, 20 km	1990–2004, MACEDA	Baseline conflict geography
Population	2002 Census	Initial market size
Population density	2002 Census	Initial settlement density
Rural population share	2002 Census	Rural-hinterland composition
Mapuche population share	1992 Census	Baseline Indigenous demographic presence
Plantation-forest density	1997 Agricultural Census	Baseline plantation-forest geography
Principal-road access	2026 MOP network snapshot	Access and connectivity
Historical title share	Pre-period TDM geography	Historically recognized claims
Capital status	Time-invariant administrative geography	Differential evolution of administrative centers

Notes: Each listed characteristic is interacted with a linear calendar-time index. Conflict history cumulates rural-only main events within 20 kilometers over 1990–2004, excluding all 2005 events. It differs from the contemporary broad-external 30-kilometer treatment. The road measure uses a 2026 network snapshot; its pre-2005 timing has not been established. Capital status is a separate time-invariant administrative indicator.

Wild-cluster and spatial-HAC inference are similar for the two headline outcomes (Table A8). The spatial-HAC calculation allows correlation across markets within 60 kilometers and serial dependence within markets through three annual lags. The 30- and 90-kilometer cutoffs yield the same qualitative conclusion.

B.2 Geographic-time fixed effects and province influence

A concern with the market-level estimates is that conflict exposure may coincide with economic shocks that vary across southern Chile over time. I therefore replace the common year fixed effects

Table A7: Formal activity under alternative baseline-trend bundles

Outcome	Market and year FE	Preferred model	Reduced trend bundle
Registered enterprises	-0.0240*** (0.0091)	-0.0112** (0.0050)	-0.0099** (0.0049)
Tax-reported sales	-0.0627*** (0.0215)	-0.0475*** (0.0146)	-0.0468*** (0.0163)
Markets	104	104	104
Market-years	1,768	1,768	1,768

Notes: The first column is the market and year fixed-effects benchmark and does not include own-market conflict. The preferred and reduced columns add own-market conflict. The reduced bundle omits only the trends for conflict exposure through 2004, baseline Mapuche share, and historical title share. Standard errors are clustered by market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Alternative inference for headline formal outcomes

Outcome	Coef.	Clustered SE	Wild p	Spatial-HAC SE	Spatial-HAC p
Registered enterprises	-0.0112**	(0.0050)	0.025	(0.0047)	0.019
Tax-reported sales	-0.0475***	(0.0146)	< 0.001	(0.0116)	< 0.001

Notes: Wild-cluster p -values use 999 repetitions and market clustering. Spatial-HAC standard errors use a 60-kilometer spatial cutoff and a three-year lag cutoff; 30- and 90-kilometer cutoffs give the same qualitative conclusion. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with region-by-year and province-by-year fixed effects, allowing annual shocks to differ across these geographies. The formal-activity coefficients remain negative in both cases. With region-by-year fixed effects, the estimates are -0.0170 for registered enterprises and -0.0526 for tax-reported sales. With province-by-year fixed effects, they are -0.0067 and -0.0398 , respectively. The enterprise estimate becomes less precise in the latter specification, but remains similar in magnitude to the main estimate.

Table A9: Formal activity with geographic-by-year fixed effects

	Registered enterprises	Tax-reported sales
Region $\times$ year fixed effects	-0.0170** (0.0071)	-0.0526*** (0.0159)
Province $\times$ year fixed effects	-0.0067 (0.0076)	-0.0398** (0.0187)
Markets	104	104
Market-years	1,768	1,768

Notes: Each cell reports the coefficient on log external conflict exposure within 30 kilometers. All specifications include market fixed effects, baseline-characteristic trends, a capital-status trend, contemporaneous own-market conflict, and market-clustered standard errors. Region-by-year and province-by-year fixed effects replace the common year fixed effects in the corresponding rows. The sample contains 104 markets and 1,768 market-years. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I also re-estimate the preferred specification after omitting each province in turn. The coefficient remains negative in every case for both outcomes. Enterprise estimates range from -0.0135 to -0.0066 , while sales estimates range from -0.0530 to -0.0366 . The aggregate association is therefore not driven by any single province in the study corridor.

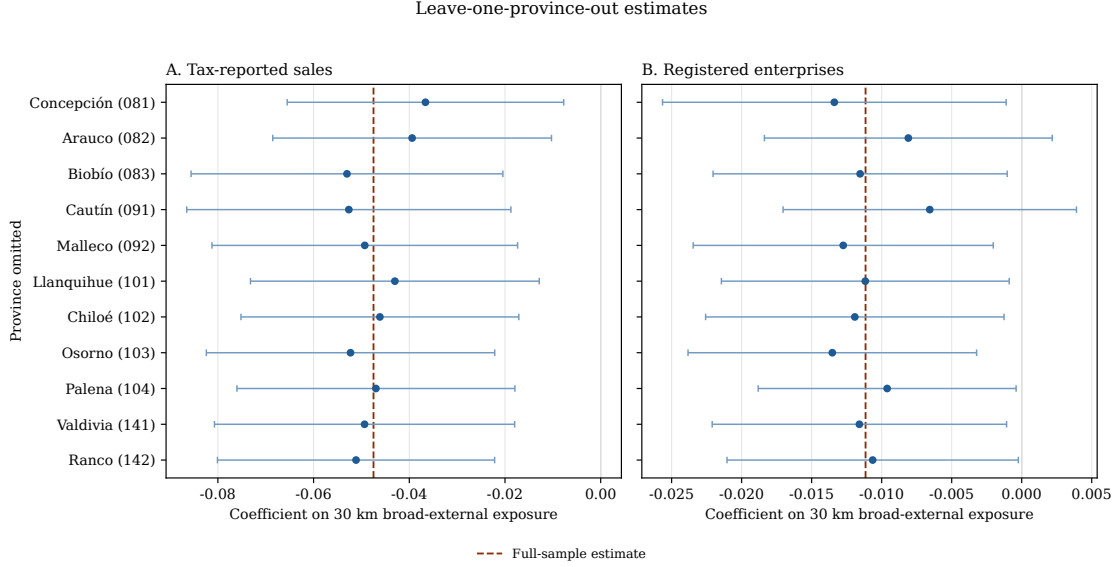


Figure A3: Leave-one-province-out estimates for formal activity. Each point reports the estimate obtained after omitting the indicated province. Vertical lines show the corresponding full-sample estimates, and horizontal bars report 95 percent confidence intervals.

B.3 Distance sensitivity

The main specification measures external conflict within 30 kilometers of each urban market. To assess whether the formal-activity results depend on this particular spatial window, Table A10 reconstructs external exposure using cumulative radii from 10 to 60 kilometers. The coefficients remain negative for both registered enterprises and tax-reported sales throughout the range. I interpret this exercise as sensitivity to the spatial window rather than as evidence for an optimal radius of influence.

B.4 Aquaculture exclusion

Los Lagos contains the largest concentration of aquaculture activity in the study corridor. Because this sector represents a large share of observed sales in the region, a sector-specific downturn could potentially contribute to the aggregate formal-activity relationship. I therefore reconstruct both SII outcomes after excluding subsector 032 (Acuicultura).

For registered enterprises, excluding aquaculture leaves the full 104-market panel unchanged and moves the coefficient only from -0.0112 to -0.0114 . Subsector-level sales are subject to statistical suppression, so the corresponding comparison must be made on the 1,066 market-years for which aquaculture sales are observed. On this common sample, the coefficient for total sales is -0.0516 ; after removing aquaculture sales it is -0.0550 . The formal-activity association is therefore not explained by the inclusion of the aquaculture sector.

Table A10: Cumulative external-conflict windows and formal activity

External window	Registered enterprises	Tax-reported sales
Within 10 km	-0.0085 (0.0075)	-0.0322 (0.0243)
Within 20 km	-0.0079 (0.0056)	-0.0337** (0.0157)
Within 30 km	-0.0112** (0.0050)	-0.0475*** (0.0146)
Within 40 km	-0.0120** (0.0049)	-0.0518*** (0.0177)
Within 50 km	-0.0095** (0.0047)	-0.0410** (0.0191)
Within 60 km	-0.0077 (0.0049)	-0.0404** (0.0182)
Markets	104	104
Market-years	1,768	1,768

Notes: Each row replaces the 30-kilometer exposure measure with log external conflict accumulated within the indicated radius. The event family, own-market conflict measure, baseline-characteristic trends, capital-status trend, fixed effects, and market clustering are otherwise unchanged. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Formal activity excluding aquaculture

Outcome	Main estimate	Excluding subsector 032	Market-years
Registered enterprises	-0.0112** (0.0050)	-0.0114** (0.0050)	1,768
Tax-reported sales	-0.0516*** (0.0186)	-0.0550*** (0.0188)	1,066

Notes: Aquaculture corresponds to SII subsector 032. The enterprise comparison uses the complete 104-market panel. Because subsector-level sales are suppressed in some municipality-years, the sales comparison uses the same observable sample before and after excluding aquaculture. All specifications otherwise reproduce the main market-level model. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 Margins of adjustment

The enterprise-stock result does not reveal whether formal activity adjusts through fewer entries, more exits, or both. I examine this distinction using SII records of first registrations and recorded terminations. First registrations capture new entries into the administrative registry, while terminations record the cessation of registered activity. The former is transformed as $\log(1 + x)$ because some markets record no new registrations in a given year.

External conflict exposure is associated with fewer first registrations. The coefficient is -0.0163 , compared with -0.0112 for the stock of registered enterprises. Recorded terminations show a much weaker pattern: the contemporaneous coefficient is 0.0029 and is imprecisely estimated. The decline in the enterprise stock therefore appears to reflect, at least in part, fewer new registrations rather than a clear increase in recorded exits.

Table A12: Enterprise stocks, first registrations, and terminations

Outcome	Transformation	Coefficient	Clustered SE
Enterprise stock	$\log(x)$	-0.0112^{**}	0.0050
First registrations	$\log(1 + x)$	-0.0163^{**}	0.0078
Current terminations	$\log(1 + x)$	0.0029	0.0145

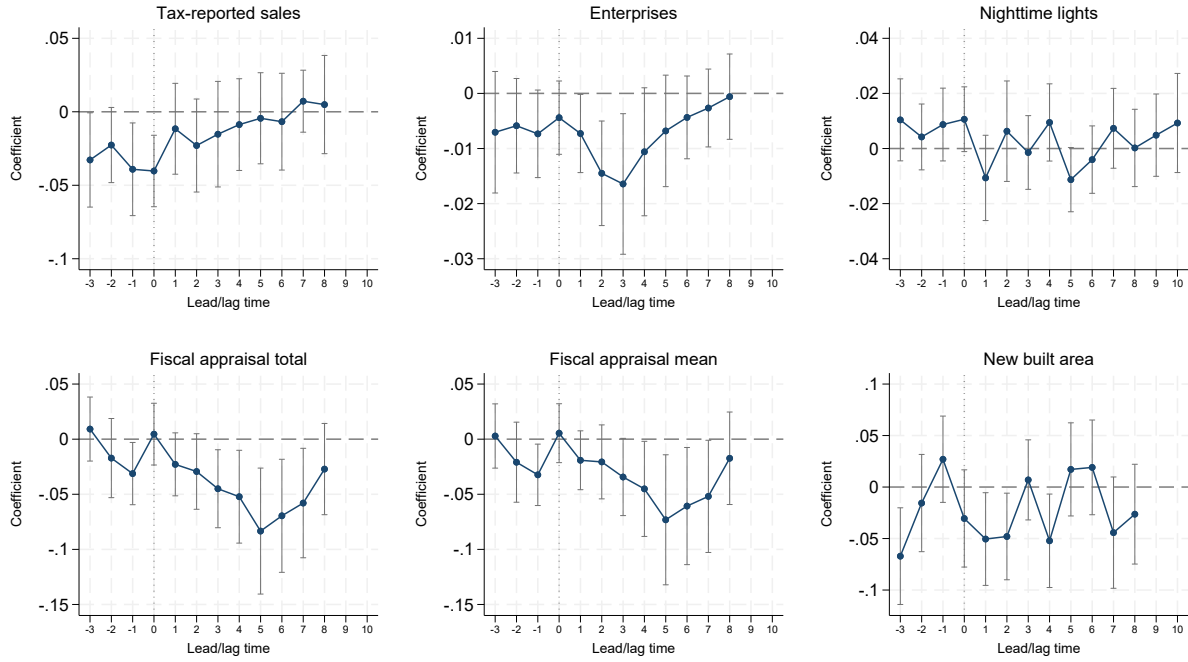
Notes: All rows use the main market-level specification and contain 1,768 market-years. The enterprise stock is measured in logs, while first registrations and terminations use $\log(1 + x)$ to retain zero-count market-years. SII termination records contain an administrative break around 2016–2017, which limits their interpretation over the full sample period. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The administrative break in the termination series prevents a clean entry-versus-exit decomposition. The evidence is nevertheless informative about the margin on which the enterprise result is most visible: lower first registrations accompany the decline in the registered enterprise stock, whereas the full-period data show little comparable increase in recorded terminations.

C Timing and identification

C.1 Continuous lead-lag results

The continuous lead-lag projection includes future, contemporaneous, and lagged values of annual external conflict exposure. For presentation, coefficients on future exposure are plotted at negative horizons. These terms capture whether current outcomes covary with exposure realized in subsequent years, conditional on the remaining exposure terms and controls. The contemporaneous and lagged coefficients describe the association with current and past exposure. Because conflict exposure is continuous and recurrent, the coefficients describe temporal patterns in the exposure–outcome relationship rather than event time around a treatment onset.



Broad external A within 30 km, conditional on annual own-market I with no distance cutoff.

Figure A4: Continuous lead-lag profiles for retained aggregate outcomes. Negative horizons denote future exposure terms, zero denotes current exposure, and positive horizons denote lagged exposure. Bars show 95 percent confidence intervals based on market-clustered standard errors.

Table A13: Summary of continuous lead-lag profiles

	(1)	(2)	(3)	(4)	(5)	(6)
	Tax-reported sales	Enter- prises	Nighttime lights	Fiscal appraisal total	Fiscal appraisal mean	New built area
Joint leads -3/-1	2.270*	1.103	1.322	2.983**	2.949**	3.889**
	p=0.085	p=0.351	p=0.271	p=0.035	p=0.036	p=0.011
Average current/lag 0-2	-0.025**	-0.008**	0.002	-0.011	-0.008	-0.042***
	(0.012)	(0.003)	(0.006)	(0.012)	(0.011)	(0.013)
Average lags 3-5	-0.010	-0.010*	-0.001	-0.057***	-0.048**	-0.010
	(0.013)	(0.005)	(0.004)	(0.020)	(0.021)	(0.012)
Average lags 6-8	0.001	-0.002	0.002	-0.049**	-0.041*	-0.023*
	(0.012)	(0.004)	(0.004)	(0.022)	(0.023)	(0.013)
Observations	1,456	1,456	1,964	936	936	1,872
Markets	104	104	104	104	104	104
Window	2005-2018	2005-2018	2000-2018	2010-2018	2010-2018	1998-2015

Notes: The joint-lead row reports the Wald statistic and its p-value for the three future-exposure coefficients. Current-and-lag rows are covariance-aware averages of the indicated current and lagged coefficients. All models use the preferred controls and market-clustered standard errors. Windows differ with outcome availability. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Major conflict escalations

The spike design isolates unusually large annual escalations in the same broad external conflict count used in the main analysis. The primary measure is an annual adaptation of the salience rule in Barrett et al. (2022). It requires a strict local peak with at least two events and an unusually high count relative to the market’s full history or recent past. Because this measure uses the following year’s count to identify a local peak and uses full-history information in two of its three thresholds, it captures isolated salience rather than a fully predetermined shock. The Past4SD comparison instead requires at least three events and a count strictly above the mean plus four sample standard deviations calculated from the preceding ten years, with all ten prior observations available.

I estimate non-normalized event-time effects with the non-absorbing estimator of de Chaisemartin and D’Haultfoeuille (2024). Event time is anchored at a market’s first within-panel change from zero to one, but the observed annual treatment path is retained after that switch. The comparison therefore contrasts the realized path with a path that remains at its initial untreated value; it does not stack later spike entries as new event-zero observations. The specification allows linear trends to vary with 2002 population, density, and rural share, the 1992 Mapuche share, historical *Títulos de Merced* exposure, 1997 plantation-forest density, and capital status. Identification requires conditional parallel trends and no anticipation for the switchers and comparison markets used at each horizon.

Figure A5 shows the central contrast. Registered enterprises change little during the placebo periods and decline progressively after a major escalation. The profile reaches -0.056 log points at five years under the Barrett-style rule and -0.070 under Past4SD. The corresponding joint placebo tests have p -values of 0.558 and 0.635, and the same-switcher estimates preserve the post-escalation pattern. Tax-reported sales are noisy and non-monotonic. Nighttime lights likewise do not reproduce the progressive negative enterprise profile. Barrett fiscal-appraisal totals remain close to zero at short horizons and fall by 0.169 log points at five years (standard error 0.070), consistent with a slower appraisal response.

Support declines with the horizon but remains broader for registered enterprises than for appraisal. At five years, 40 Barrett and 30 Past4SD enterprise switchers contribute to the effect; 5.0 and 6.7 percent of their supported paths, respectively, contain another spike by that horizon. The appraisal window begins in 2010, leaving 16 Barrett and 10 Past4SD switchers in the five-year same-switcher comparison. These counts reinforce the use of the enterprise profile as the central result and the Barrett appraisal pattern as suggestive evidence only.

Table A15 shows that prior conflict history modestly predicts entry into both spike definitions, but prior declines in enterprises, sales, and nighttime lights do not systematically predict entry. For Past4SD, the only precise one-year enterprise predictor is positive enterprise growth rather than deterioration. This reduces concern about a simple reverse-timing account based on recent economic decline, but it does not establish that spike entry is exogenous.

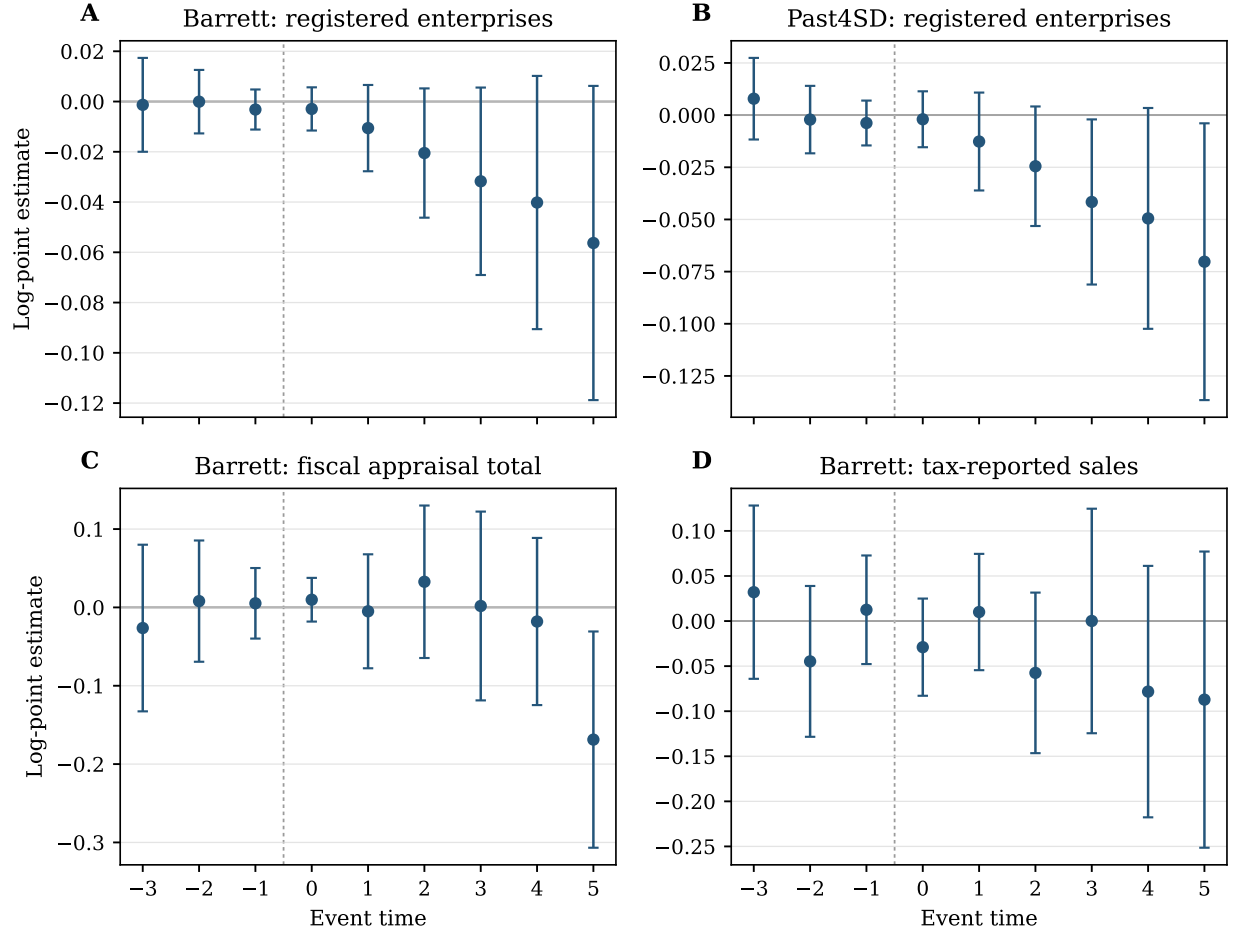


Figure A5: Major conflict escalations and urban-market outcomes. Points report non-normalized effects from the de Chaisemartin–D’Haultfoeuille estimator; vertical bars are 95 percent confidence intervals. The annual treatment remains non-absorbing after the first within-panel switch. Estimates at negative event times are placebo estimators, and the dotted vertical line separates them from post-switch effects. All specifications include the baseline-characteristic trend adjustment described in the text. Inference uses the estimator’s market-level variance.

Table A14: Dynamic effects following major conflict escalations

Outcome	Definition	t	+1	+2	+3	+5	Placebo p
Registered enterprises	Barrett	-0.003 (0.004)	-0.011 (0.009)	-0.020 (0.013)	-0.032 (0.019)	-0.056 (0.032)	0.558
	Past4SD	-0.002 (0.007)	-0.013 (0.012)	-0.024 (0.015)	-0.042 (0.020)	-0.070 (0.034)	0.635
Tax-reported sales	Barrett	-0.029 (0.028)	0.010 (0.033)	-0.057 (0.045)	0.000 (0.064)	-0.087 (0.084)	0.108
	Past4SD	-0.016 (0.026)	-0.004 (0.036)	-0.033 (0.044)	0.043 (0.071)	-0.087 (0.094)	0.073
Urban nighttime lights	Barrett	0.049 (0.016)	-0.032 (0.020)	-0.020 (0.027)	0.007 (0.037)	0.021 (0.047)	0.090
	Past4SD	0.054 (0.016)	0.005 (0.022)	0.011 (0.029)	0.043 (0.041)	0.074 (0.057)	0.006
Fiscal appraisal total	Barrett	0.010 (0.014)	-0.005 (0.037)	0.033 (0.050)	0.002 (0.061)	-0.169 (0.070)	0.651
	Past4SD	-0.003 (0.025)	-0.034 (0.040)	-0.055 (0.055)	-0.117 (0.110)	-0.240 (0.133)	0.005
Fiscal appraisal mean	Barrett	0.019 (0.014)	0.010 (0.037)	0.051 (0.050)	0.019 (0.059)	-0.100 (0.071)	0.297
	Past4SD	-0.001 (0.024)	-0.025 (0.040)	-0.042 (0.054)	-0.092 (0.103)	-0.196 (0.122)	0.017

Notes: Cells report non-normalized DCDH estimates, with standard errors in parentheses. The final column reports the joint p -value for the three placebo estimators. Barrett is the annual local-salience composite; Past4SD uses only the preceding ten years and requires at least three events. The annual treatment path remains non-absorbing after the first switch. All models use the baseline-characteristic trend adjustment and market-level inference. Past4SD nighttime-light and fiscal-appraisal placebo tests reject at conventional levels, so those estimates are reported for completeness but are not used as supporting identification evidence.

Table A15: Do recent economic changes predict entry into a major conflict escalation?

	Barrett	Past4SD
<i>Panel A. One-year changes before entry</i>		
	0.095	0.136**
Enterprise growth, $t - 1$	(0.062)	(0.064)
	-0.011	-0.029
Sales growth, $t - 1$	(0.017)	(0.020)
<i>Panel B. Earlier changes</i>		
	0.005	-0.021
Enterprise growth, $t - 2$	(0.071)	(0.070)
	0.046	0.014
Enterprise growth, $t - 3$	(0.059)	(0.042)
	0.030	0.016
Sales growth, $t - 2$	(0.018)	(0.015)
	-0.013	-0.002
Sales growth, $t - 3$	(0.015)	(0.011)
<i>Panel C. Predictive content of prior conflict</i>		
Year-FE R^2	0.044	0.046
Conflict-history R^2	0.068	0.061
Incremental R^2	0.024	0.015
Number of spike entries	76	47

Notes: The dependent variable is entry into a spike from an at-risk non-spike year. Economic predictors are dated strictly before spike entry. Panel A includes the two one-year changes jointly, while Panel B enters each earlier change separately. The economic-change regressions include market and year fixed effects and controls for recent conflict history. Standard errors, reported in parentheses, are clustered by market. The exercise tests whether recent economic deterioration systematically precedes escalation; it does not establish exogeneity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Reverse-predictive evidence

Table A16 asks whether formal activity predicts subsequent external conflict. One-year changes in enterprises and sales do not predict exposure one or two years later. Lower levels do predict later exposure, including after conditioning on current exposure and its first two lags. This weakens a simple short-run feedback story from recent deterioration to subsequent conflict, but leaves slow-moving confounding, conflict history, and longer-run feedback unresolved.

Table A16: Do formal-activity levels or recent changes predict future external conflict?

Specification	Predictor	Horizon	Coef.	SE	p-value	N
Level	Enterprises level	t+1	-0.151	0.094	0.111	1,560
Level	SII sales level	t+1	-0.103	0.044	0.020	1,560
Level	Enterprises level	t+2	-0.148	0.082	0.074	1,456
Level	SII sales level	t+2	-0.066	0.035	0.061	1,456
One-year change	Enterprises change	t+1	0.084	0.137	0.544	1,560
One-year change	SII sales change	t+1	-0.038	0.037	0.295	1,560
One-year change	Enterprises change	t+2	-0.034	0.109	0.757	1,456
One-year change	SII sales change	t+2	0.013	0.037	0.731	1,456
Level + conflict history	Enterprises level	t+1	-0.130	0.079	0.103	1,560
Level + conflict history	SII sales level	t+1	-0.085	0.039	0.029	1,560
Level + conflict history	Enterprises level	t+2	-0.153	0.085	0.075	1,456
Level + conflict history	SII sales level	t+2	-0.069	0.036	0.057	1,456
Change + conflict history	Enterprises change	t+1	0.061	0.126	0.629	1,560
Change + conflict history	SII sales change	t+1	-0.041	0.040	0.299	1,560
Change + conflict history	Enterprises change	t+2	-0.035	0.111	0.750	1,456
Change + conflict history	SII sales change	t+2	0.015	0.038	0.693	1,456

Notes: Each cell is a coefficient from a projection of future external conflict exposure on the indicated enterprise or sales measure. The history-conditioned columns additionally control for current exposure and its first two annual lags. All models use market and year fixed effects, the relevant baseline-characteristic trends, and market-clustered standard errors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Within-market spatial incidence

The aggregate estimates describe how formal activity varies across urban markets over time. I complement them with a finer spatial comparison that asks whether locations within the same urban market are differently exposed to nearby external conflict. For each *manzana*, exposure is constructed from its spatial proximity to external conflict events within the same 30-kilometer window used in the aggregate analysis.

The specification includes *manzana* fixed effects and market-by-year fixed effects. The former absorb persistent differences across blocks, while the latter absorb all shocks shared by locations in the same urban market and year. Variation therefore comes from differences in exposure across *manzanas* within a given market-year. The coefficients describe relative spatial incidence inside the urban footprint rather than changes in aggregate market-level activity.

Mean fiscal appraisal is positive in this comparison, while nighttime lights, total fiscal appraisal,

newly built area, and commercial land use do not reveal a common negative within-market gradient. The exercise therefore does not show a broad pattern in which more exposed *manzanas* decline relative to other locations in the same market-year.

Table A17: Within-market spatial incidence at the 30-kilometer external window

	Activity proxy	Assessed stock		Cadastre composition	Built-form flow
	(1)	(2)	(3)	(4)	(5)
	VIIRS urban lights	Fiscal appraisal sum	Fiscal appraisal mean	Commerce-use share (pp)	WSF new
External conflict exposure	-0.006 (0.021)	0.040* (0.023)	0.042** (0.019)	0.033 (0.030)	0.001 (0.003)
Observations	314,298	393,800	393,800	331,991	209,532
Manzanas	34,922	34,050	34,050	34,050	34,922
Markets	104	104	104	104	104
Window	2013–2021	2010–2021	2010–2021	2012–2021	2010–2015

Notes: Each column includes *manzana* fixed effects and market-by-year fixed effects. The latter absorb own-market conflict and all shocks common to locations within the same urban market and year. Standard errors are clustered by market. Outcome windows differ with data availability. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Municipal-versus-urban interpretation

E.1 Registered legal entities

The SII outcomes used in the main analysis are assigned through municipal tax addresses and therefore do not mechanically locate economic activity inside the fixed urban footprints. I first examine this measurement issue using registered legal entities (*personas jurídicas*) whose historical headquarters can be matched with high confidence to cadastral parcels.

The linked sample is necessarily selected, and a headquarters address need not coincide with the location of production or sales. Nevertheless, the association with external conflict remains negative in the parcel-linked sample and among legal entities whose historical headquarters are located inside the fixed urban footprints. This provides spatial evidence that the main formal-activity pattern is not confined to businesses that can only be assigned at the municipality level, while stopping short of identifying the urban share of all economic activity.

E.2 Rural production linkages

A second concern is that municipality-level SII outcomes could primarily reflect businesses tied to the surrounding rural economy rather than activity associated with urban markets. I examine this possibility using predetermined sectoral linkages from the Chilean input-output matrices (*Matriz Insumo-Producto*, MIP) published by the Banco Central de Chile.

For each sector, I construct two measures of its direct connection to rural production. The rural input linkage is the share of the sector’s domestic intermediate inputs purchased from agriculture,

Table A18: External conflict and registered legal entities with geocoded headquarters

Outcome	Transformation	Coefficient	Clustered SE	Zero market-years
Registered enterprises	$\log(x)$	-0.0112**	0.0050	0
Total registered legal entities	$\log(x)$	-0.0105	0.0086	0
Temporally assigned legal entities	$\log(x)$	-0.0157*	0.0094	0
Cadastral-classified legal entities	$\log(1 + x)$	-0.0182	0.0202	109
Cadastral urban legal entities	$\log(1 + x)$	-0.0282	0.0193	152

Notes: Historical headquarters are matched to cadastral parcels using the administrative address records. The linked samples are selected and headquarters locations need not identify production or sales locations. Zero-count market-years are retained. Standard errors are clustered by market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

fruit growing, livestock, and forestry. The rural customer linkage is the share of the sector's domestic intermediate sales sold to those same rural activities. Fishing and aquaculture are excluded from the rural group. The 2003 MIP is the preferred benchmark because it is the closest pre-period matrix to the 2005–2021 estimation window; the 1996 MIP provides an earlier benchmark. The linkage measures are standardized across the ten comparable non-rural sectors retained in the preferred sector panel.

Let L_s denote either standardized linkage for sector s . I estimate

$$Y_{mst} = \beta (D_{mt} \times L_s) + \rho (I_{mt} \times L_s) + \alpha_{ms} + \gamma_{st} + \delta_{mt} + \varepsilon_{mst}, \quad (4)$$

where D_{mt} is external conflict exposure and I_{mt} is own-market conflict. Market-by-sector fixed effects absorb persistent differences in local sector composition, sector-by-year fixed effects absorb national sectoral shocks, and market-by-year fixed effects absorb all shocks common to sectors in the same market and year. The coefficient β therefore asks whether the formal-activity association is systematically different in sectors that are more structurally connected to rural production.

Table A19: External conflict and predetermined rural sectoral linkages

Benchmark	Linkage	Sales interaction	Enterprise interaction
2003	Customer	-0.0097 (0.0268)	0.0054 (0.0046)
2003	Input	0.0040 (0.0352)	-0.0045 (0.0031)
1996	Customer	-0.0062 (0.0357)	0.0036 (0.0048)
1996	Input	0.0079 (0.0335)	-0.0037 (0.0032)

Notes: Input and customer linkages are constructed from the Banco Central de Chile input-output matrices and standardized across the ten sectors in the preferred sample. The 2003 matrix is the main pre-period benchmark and the 1996 matrix provides an earlier benchmark. Each coefficient is the interaction between log external conflict exposure within 30 kilometers and the indicated standardized sectoral linkage. Specifications also include the corresponding interaction with own-market conflict and absorb market-by-sector, sector-by-year, and market-by-year fixed effects. Standard errors are clustered by market. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction estimates are small and do not display a systematic negative gradient for either sales or enterprises. The conclusion is similar using the 2003 input and customer linkages and the earlier 1996 benchmark. The aggregate formal-activity association therefore does not appear to be disproportionately concentrated in sectors with stronger structural links to rural production.

This exercise does not locate SII sales or enterprises inside the urban footprint. Rather, it complements the headquarters evidence above: the cadastre linkage provides selected but directly geocoded evidence inside the urban footprint, while the input-output exercise shows that the aggregate result is not systematically stronger in sectors whose production relationships are more rural-oriented.

F Historical heterogeneity

I examine whether the association with contemporary external conflict varies with two historical characteristics of the receiving market. The first is territorial: the preferred measure is the share of terrestrial land in the external 0–30 km catchment covered by historical *Títulos de Merced* (TDM). The second is demographic: the Mapuche share of the urban population in 1992, reconstructed from individual census records and harmonized to the fixed urban-market geography. The two measures capture related but distinct dimensions of the historical setting and enter jointly in the specifications.

Table A20 reports the preferred 30-km results. The estimates provide no evidence that the negative formal-activity association becomes systematically stronger either in markets surrounded by more historically titled Mapuche territory or in markets with a larger baseline Mapuche population. The TDM interaction is close to zero for registered enterprises and positive but imprecisely estimated for tax-reported sales. The Mapuche-share interactions likewise show no systematic negative gradient. Figure A6 shows that the TDM result is similar when the surrounding historical geography is defined using 10-, 20-, 30-, or 50-km catchments. These results do not imply that Indigenous presence or historical territorial claims are economically irrelevant; rather, they provide no evidence that either dimension mechanically generates the negative formal-activity gradient associated with contemporary conflict exposure.

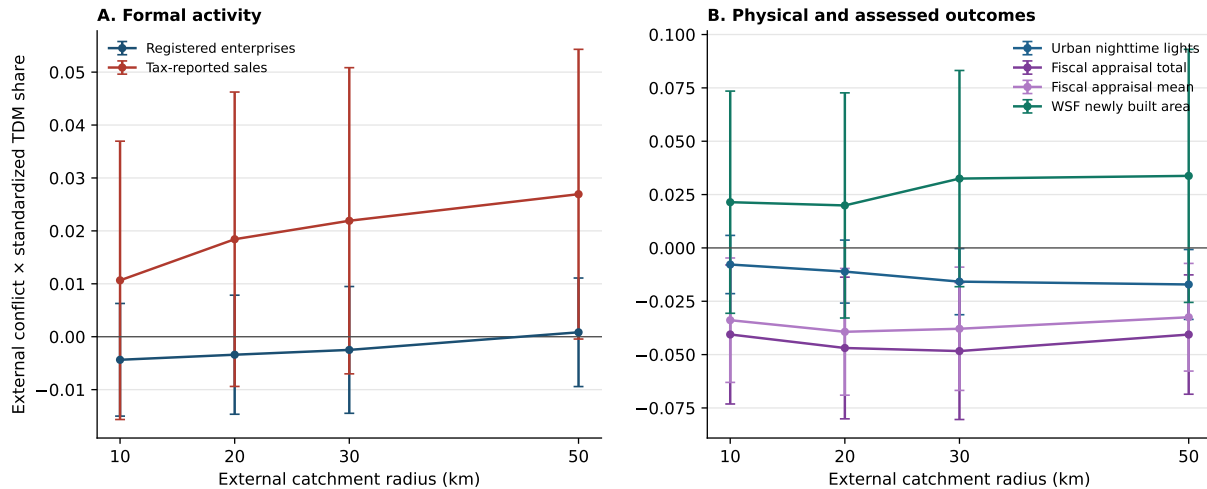
The fiscal-appraisal outcomes display a different pattern. Their interactions with TDM exposure are negative across the four catchments and remain negative when external- and own-market conflict slopes are allowed to vary by region. No comparable pattern appears with baseline Mapuche population share. Fiscal appraisal is a slowly updated assessed property stock rather than a market price, so this evidence does not identify a property-rights mechanism. It does, however, indicate that the association between conflict exposure and assessed property stocks is more negative in markets surrounded by greater historical TDM coverage. I treat this as secondary evidence, particularly because TDM coverage is geographically concentrated: at the preferred 30-km radius, 42 of 104 markets have no TDM territory in the external catchment, and much of the positive variation is concentrated in Araucanía.

Table A20: Historical territorial exposure, baseline Mapuche presence, and urban-market outcomes

Outcome	TDM share, external 0–30 km			1992 Mapuche urban-population share		
	β (SE)	Wild p	BH q	β (SE)	Wild p	BH q
<i>Panel A. Formal activity</i>						
Registered enterprises	-.002 (.006)	.703	.733	.010 (.011)	.400	.545
Tax-reported sales	.022 (.015)	.123	.246	-.013 (.022)	.588	.641
<i>Panel B. Physical and assessed outcomes</i>						
Urban nighttime lights	-.016 (.008)	.067	.146	.007 (.011)	.564	.641
Fiscal appraisal total	-.048 (.016)	.005	.048	.037 (.018)	.051	.201
Fiscal appraisal mean	-.038 (.015)	.014	.060	.025 (.017)	.155	.310
WSF newly built area	.032 (.026)	.266	.422	-.030 (.033)	.408	.545

Notes: The table reports interactions between external conflict exposure and two standardized historical moderators, estimated jointly. TDM exposure is the share of terrestrial land in the receiving market's external 0–30 km catchment covered by historical *Títulos de Merced*. Mapuche population share is reconstructed from individual records of the 1992 Census and measures the Mapuche share of the historical urban population harmonized to the fixed urban-market geography. The joint sample contains 102 markets; Curaco de Vélez and Palena have no urban population records in the 1992 census geography. All specifications include market and year fixed effects, outcome-specific baseline-characteristic trends, a capital-status trend, own-market conflict and the corresponding moderator interactions. Standard errors are clustered by market and wild-cluster p -values are reported. TDM q -values adjust the 24 pre-specified radius-by-outcome tests; Mapuche q -values use the pre-specified Mapuche interaction family.

Figure A6: Historical TDM exposure across alternative catchment boundaries



Notes: Points report interactions between external conflict exposure and standardized TDM land share; bars show 95 percent confidence intervals based on market-clustered standard errors. The 10-, 20-, 30-, and 50-km catchments are alternative boundaries for the same measure of surrounding historical territorial exposure and are highly correlated.

G Institutional response: CONADI

CONADI records provide evidence on formal land-acquisition activity in the same territorial setting. Article 20(a) and 20(b) acquisitions are aggregated to market-years using the official municipality codes and the municipality-market crosswalk. The source reports an acquisition year, although the precise administrative stage represented by that date is not documented. The two outcomes are the log of one plus the number of recorded acquisitions and the log of one plus reported acquired hectares.

Greater external conflict exposure is followed by more recorded CONADI acquisition activity over the subsequent two years, with the largest coefficients at one- and two-year lags. This documents continued formal land-acquisition activity in conflict-exposed territories; it does not identify the effect of CONADI acquisitions on conflict or local economic activity.

Table A21: External conflict exposure and recorded CONADI land acquisition

Outcome	Static	D_t	D_{t-1}	D_{t-2}	Joint p	Cumulative
Acquisition events	0.0284 (0.0284)	0.0086 (0.0300)	0.0643** (0.0323)	0.1054*** (0.0279)	< 0.001	0.1783*** (0.0402)
Reported hectares	0.0811 (0.0724)	0.0449 (0.0770)	0.1009 (0.0892)	0.2682*** (0.0781)	< 0.001	0.4140*** (0.1093)

Notes: The static model uses contemporaneous external conflict exposure. The distributed-lag model includes exposure at t , $t - 1$, and $t - 2$ jointly. The cumulative column reports the sum of these three coefficients, with its covariance-adjusted standard error. All specifications use the preferred 104-market, 2005–2021 specification and market-clustered standard errors. CONADI outcomes are transformed as $\log(1 + x)$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.